ASSIGNMENT 2 (COMBINED) | CONTROL SYSTEMS + DIGITAL SIGNAL PROCESSING

Topic 12

Electronic Stability Control (ESC)

Engineering Systems Analysis of a Commercial Product (Automotive)

Alan A Samkutty

Class: S5 ECA

Sensors → DSP → Controller → Actuator

1. System Overview

What it is

- ESC (also sold as ESP / VSC / DSC) is an active-safety system that detects when a car is **not going where the driver is steering** and corrects it by braking individual wheels and, if needed, reducing engine torque.
- Prevents **oversteer** (rear slides out) and **understeer** (front plows wide) on slippery roads, in sudden evasive manoeuvres and on split-friction surfaces.
- Fitted on production cars by suppliers such as Bosch, Continental and ZF; it is mandatory on new passenger cars in many regions (e.g. EU, US, India for new models phased in).
- Built on top of ABS and traction control: it reuses the same hydraulic modulator and wheel-speed sensors.

Key hardware

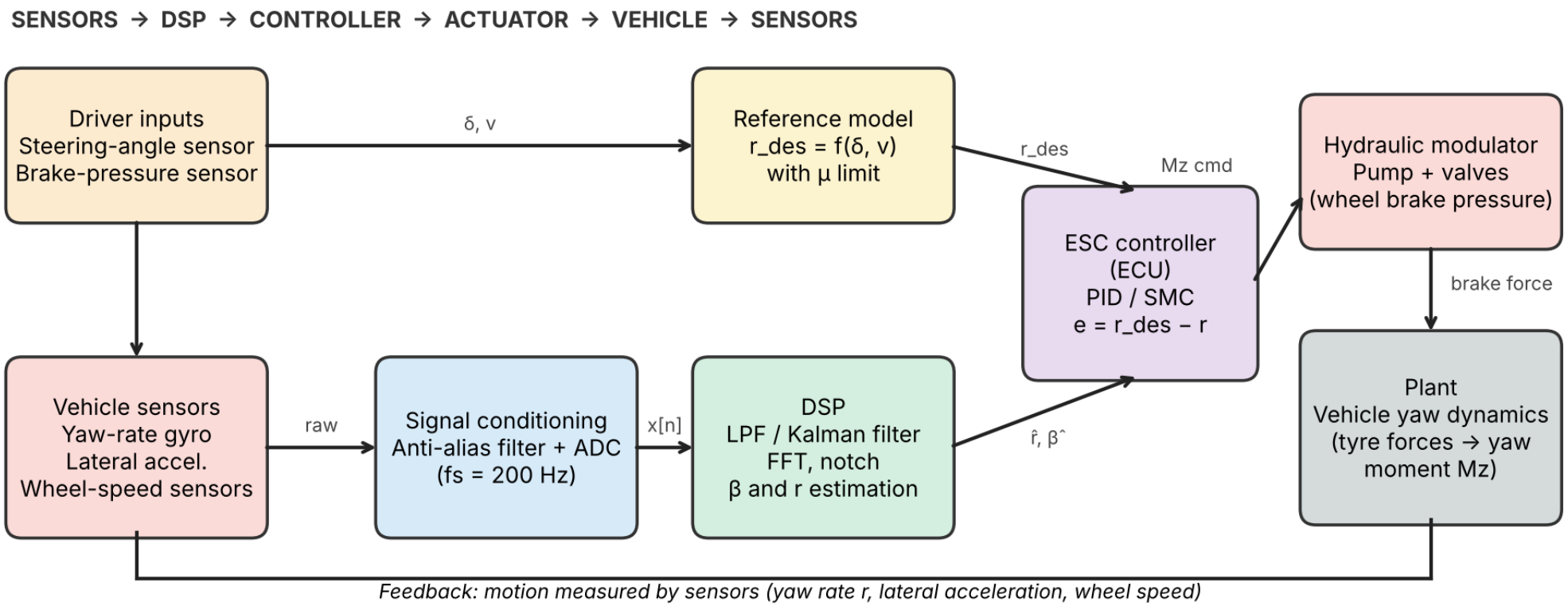
- **Sensors:** steering-wheel angle, yaw-rate gyro, lateral accelerometer, four wheel-speed sensors, brake-master-cylinder pressure.
- **ECU:** a microcontroller running the estimator and control law at a few ms sample time.
- **Actuator:** hydraulic unit (pump + solenoid valves) that applies brake pressure to any single wheel, plus engine-torque request over CAN.

How it works in one line

The ECU compares the **driver's intended yaw rate** (from steering angle and speed) with the **measured yaw rate**. The difference drives a corrective yaw moment created by braking one wheel.

Why Control + DSP together: the yaw-rate feedback loop (control) is only as good as the filtered sensor signal (DSP) it receives. Noise or lag in that signal directly degrades braking commands.

2. Block Diagram: sensor → DSP → controller → actuator



Signal flow: the gyro, accelerometer and wheel-speed sensors give raw signals; the DSP stage filters them and estimates yaw rate r and sideslip β; the controller compares r with the reference r_des computed from steering angle δ and speed v, and commands a yaw moment M_z to the hydraulic modulator, which brakes a single wheel. The vehicle's motion is measured again, closing the loop.

3. Control System

a. Plant	The vehicle's lateral/yaw dynamics. Single-track (bicycle) model: $\mathbf{m(dv_y/dt + v \cdot r) = F_{yf} + F_{yr}}$, $\mathbf{I_z \cdot dr/dt = aF_{yf} - bF_{yr} + M_z}$. Tyre forces saturate near the friction limit ($\mu \cdot F_z$), so the plant is **nonlinear**; the input is the corrective yaw moment M_z, the output is the yaw rate r (and sideslip β).
b. Controller	Hierarchical. **Upper level** computes the desired yaw rate $\mathbf{r_des = v\delta / (L + K_{us}v^2)}$, limited to about $0.85 \cdot \mu \cdot g/v$, and a required yaw moment from the error e = r_des – r using PI / PID (sliding-mode and MPC variants are used in research). **Lower level** converts M_z into brake pressure on the selected wheel (front outer wheel to counter oversteer, rear inner wheel to counter understeer) and an engine-torque cut.
c. Stability	Near the linear range the yaw dynamics are stable at all speeds for an understeering car ($K_{us} > 0$); an oversteering car becomes unstable above its critical speed, which ESC is there to prevent. Loop stability is kept by limiting loop bandwidth, anti-windup on the integrator, saturation of M_z, and by keeping the sensor-filter delay small compared with the loop time constant. Lyapunov / sliding-mode methods give formal guarantees.
d. Feedback	Negative feedback on **yaw rate** (primary) and **sideslip angle** (secondary, estimated, since it cannot be measured cheaply). Feedback signal = DSP-conditioned gyro/accelerometer data; the reference comes from the steering sensor (feed-forward from the driver).

4. Digital Signal Processing

a. Sampling / ADC	Gyro and accelerometer outputs pass an analog anti-alias filter and are sampled (typically 100–500 Hz, 12–16 bit ADC / sensor ASIC). Vehicle yaw motion is below ~5 Hz, so a few hundred Hz satisfies Nyquist with large margin; wheel-speed sensors give pulse trains whose period is measured with a timer.
b. Filtering	Digital low-pass (e.g. 2nd-order Butterworth IIR, $f_c \approx 5–10$ Hz) removes noise and engine/road vibration. Offset (bias) of the gyro is removed by a slow high-pass or estimated online. Kalman / complementary filters fuse gyro, lateral acceleration and wheel-speed data into the yaw-rate and sideslip estimate.
c. FFT / Spectral analysis	The FFT shows where signal and interference sit: manoeuvre energy lies below ~5 Hz while vibration lines and noise appear at higher frequency. It guides the cut-off choice and notch placement, and is used to check the sensor for faults (e.g. a resonant peak).
d. Feature extraction	Yaw rate r, sideslip angle β, vehicle reference speed (from the four wheel speeds), wheel slip, road-friction estimate, and error $e = r_{des} - r$. Plausibility checks compare the yaw-rate sensor against lateral acceleration and steering to detect sensor faults.
e. Noise reduction	Low-pass filtering, notch filtering of vibration, median filtering of pulse glitches and Kalman estimation. Trade-off: stronger filtering means **more phase lag** in the feedback loop.

Link between Control (3) and DSP (4)

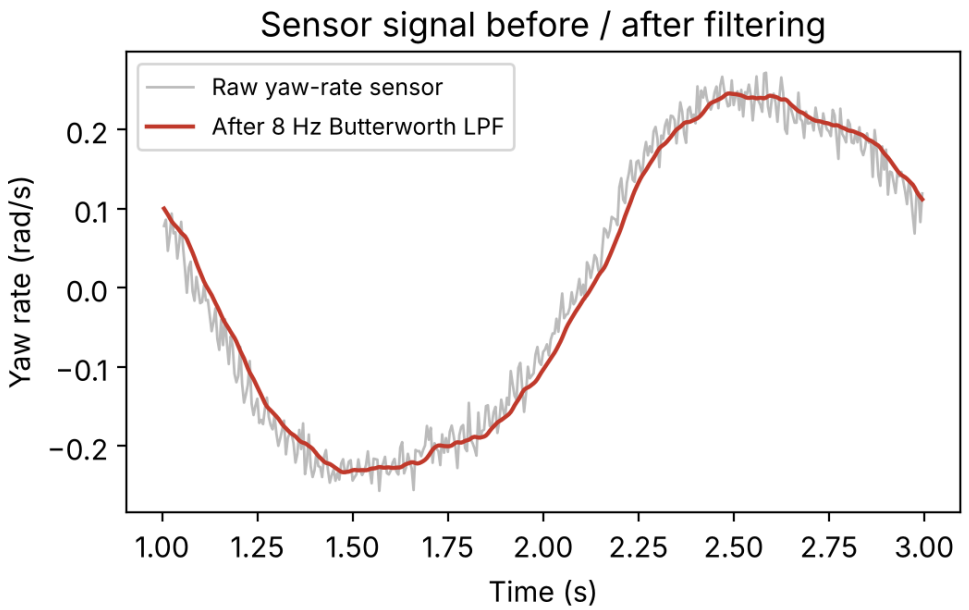
Which DSP signal becomes the feedback input?

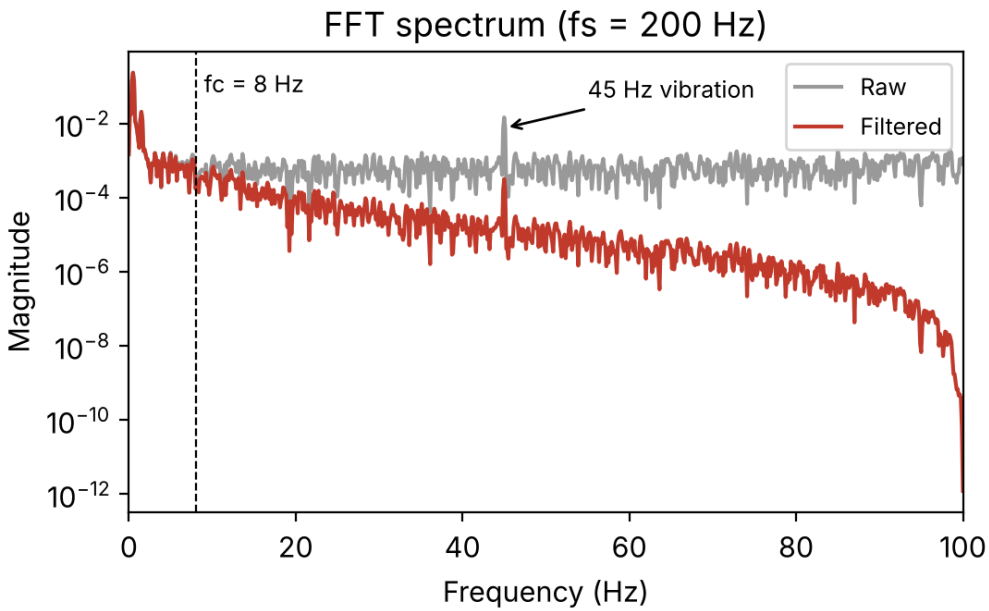
- The **filtered yaw rate $r_f[n]$** (and the estimated sideslip β_{est}) leaves the DSP stage and enters the controller as the measured output in e[n] = r_des[n] – r_f[n].
- The steering angle δ and speed v (also DSP-processed) form the reference r_des.

How controller performance depends on the DSP quality

- **Noise:** with the proportional gain K_p, sensor noise appears directly on M_z (output = K_p·noise). Un-filtered noise makes the valves chatter, which is audible, wears the hydraulics and wastes pump energy.
- **Delay:** a low-pass filter adds phase lag $\phi(f_c)$. That reduces the phase margin and slows the reaction; too low a cut-off lets the car slide further before braking begins.
- **Sampling:** the sample rate must keep the loop delay small versus the yaw time constant; aliasing folds vibration back into the signal band.
- **Bias / drift:** an uncorrected gyro offset becomes a permanent yaw-rate error that the integrator tries to remove, causing false interventions.
- **Design rule used here:** choose f_c well above the manoeuvre bandwidth (~1–2 Hz) but well below the vibration/noise band, so noise falls strongly while the added lag stays small.

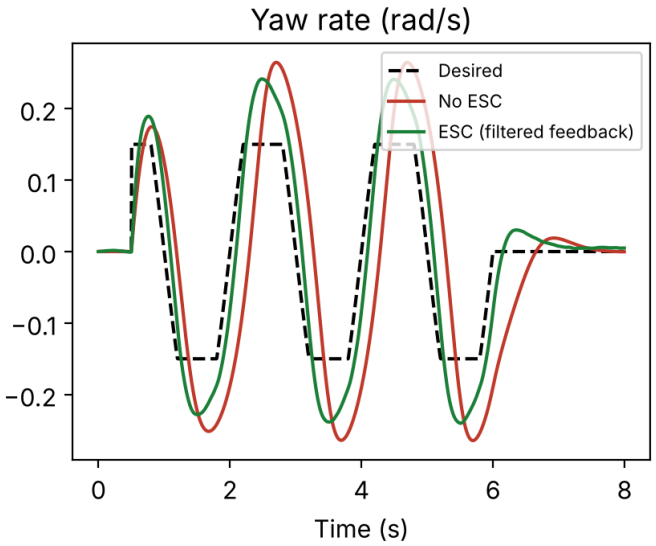
5. Simulation (Python): DSP stage

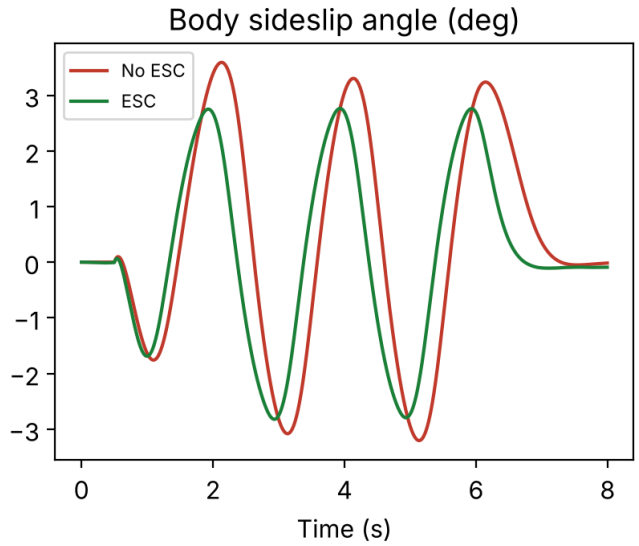


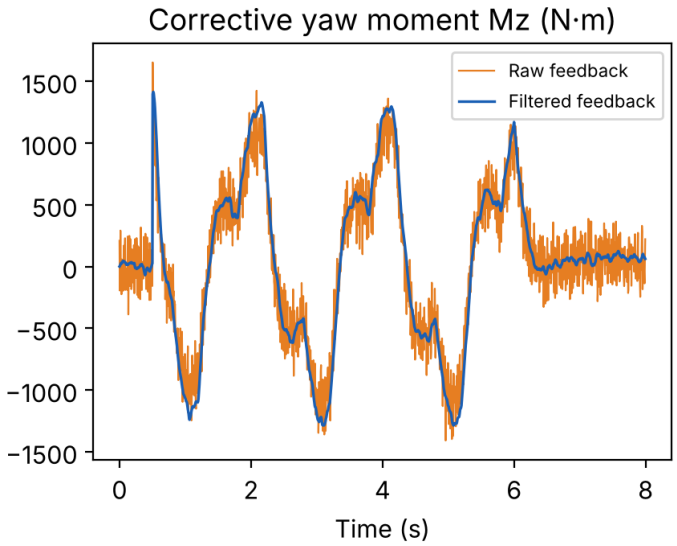


Setup: NumPy / SciPy / Matplotlib. Yaw-rate sensor modelled as true r + white noise (σ = 0.012 rad/s) + 45 Hz vibration line, sampled at 200 Hz, filtered with an 8 Hz 2nd-order Butterworth low-pass.
Result: the time plot shows the noise removed while the manoeuvre shape is kept. The FFT shows the 45 Hz vibration peak falls by about 40 dB after filtering and broadband noise above 20 Hz is greatly reduced; the useful content below ~5 Hz is unchanged.

5. Simulation (Python): closed-loop ESC







Case	RMS yaw-rate error (rad/s)	Peak sideslip (deg)	M_z chatter (RMS step, N·m)
No ESC	0.114	3.59	–
ESC, raw sensor fed back	0.057	2.73	204
ESC, filtered sensor fed back	0.057	2.82	39

Model: single-track vehicle (m = 1500 kg, I_z = 2500 kg·m², v = 25 m/s, μ = 0.45, saturating tanh tyre model), 0.5 Hz sinusoidal steering for 6 s, PI yaw-rate controller at 200 Hz applying M_z (limited to ±6 kN·m). **Findings:** ESC halves the yaw-rate tracking error and reduces peak sideslip. Feeding back the filtered signal keeps the same accuracy but cuts the controller output chatter about 5×, the benefit of DSP for the control loop. (A simplified teaching model, not a calibrated vehicle.)

6. Future Improvements (AI/ML and Emerging Technology)

AI / ML in the loop

- **Reinforcement-learning control:** RL (DDPG / TD3) can tune controller gains or distribute torque adaptively across road friction and speed, without an exact tyre model.
- **Neural-network state estimation:** a neural network plus an unscented Kalman filter gives better sideslip estimates from low-cost sensors.
- **Edge AI / TinyML:** run these estimators on the ECU microcontroller in real time with small quantised models.

Other emerging technology

- **Digital twin** of the vehicle for online parameter (mass, tyre, μ) adaptation and predictive diagnosis.
- **Sensor fusion** with GNSS/IMU, camera and radar for road-friction and sideslip estimation.
- **Torque vectoring in EVs:** in-wheel / four-motor drives apply yaw moment faster and more precisely than hydraulic brakes.
- **V2X connectivity** (5G): warn of slippery patches ahead so ESC can pre-adapt.
- **Functional safety:** ISO 26262 / SOTIF validation will be needed for any learning-based component.

7. Recent Research Related to ESC

Paper	Relevance to ESC
R. Jafari, P. Sarhadi, S. Khalil, A. Paykani, P. Asef, “A TD3-based reinforcement learning algorithm with curriculum learning for adaptive yaw control in all-wheel-drive electric vehicles,” *IEEE Access*, vol. 13, pp. 127150–127169, 2025.	Model-free RL yaw control that reduces yaw-rate error and sideslip across manoeuvres and road conditions.
R. Jafari et al., “Optimal torque allocation for all-wheel-drive electric vehicles using a reinforcement learning algorithm,” *2024 13th Mediterranean Conf. on Embedded Computing (MECO)*, IEEE, 2024. DOI 10.1109/MECO62516.2024.10577860.	LQR high-level yaw moment plus DDPG torque allocation, better dynamic stability than even torque split.
C. Zhang, Y. Feng, J. Wang, P. Gao, P. Qin, “Vehicle sideslip angle estimation based on radial basis neural network and unscented Kalman filter algorithm,” *Actuators*, vol. 12, no. 10, p. 371, 2023. DOI 10.3390/act12100371.	A neural “pseudo slip angle” fed into a Kalman filter for low-cost sideslip estimation (DSP + ML link).
J. E. Lee, B. W. Kim, “Improving direct yaw-moment control via neural-network-based non-singular fast terminal sliding mode control for electric vehicles,” *Sensors*, vol. 24, no. 13, p. 4079, 2024. DOI 10.3390/s24134079.	Sliding-mode DYC with an RBF neural network, with Lyapunov stability proof.
“A hierarchical lateral stability control strategy of distributed drive electric vehicles based on extended Kalman filter and integral terminal sliding mode control,” *PLOS ONE*, 2026. DOI 10.1371/journal.pone.0341354.	EKF state estimation (upper layer) feeding a sliding-mode yaw-stability controller.

Trend: research is moving from fixed-gain control with Kalman estimation towards learning-based estimation and control, especially for electric vehicles with independently driven wheels.